

HASSAN ELSEOUDY

Senior Software Engineer | Berlin, Germany

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Summary

Senior software engineer with 8+ years designing distributed backend systems in Java, Kotlin, and Python. At Zalando, designs and leads cross-team platforms for supplier finance and logistics, and drives platform modernization and AI-assisted engineering. Internal speaker, Gen AI Labs trainer, and startup founder.

Experience

Senior Software Engineer

Sep 2022 – Present

Zalando — Procure-to-Pay Engineering

Berlin, Germany

- **Designed the architecture and led delivery** of a **supplier-chargeback platform** that automatically bills suppliers for contract violations; its reusable design scaled to **seven chargeback types** generating **€33M+ a year** (€67M+ since 2025) across **1,200+ suppliers** and saved **~1.5 FTEs** of manual work.
- Co-authored the technical designs for a **supplier self-service tool** spanning **six services** and for clarification workflows covering **~500 suppliers**; the self-service launch saved or avoided **~3.5 FTEs** and cut delivery delays by **1.5 days**.
- Developed and operated the delivery-notice system suppliers use to announce shipments (**1.5M+ notices**, a **250M+ row** database, **99.99% availability**, **540K+** submitted by **2,300+ suppliers** since 2022) and its integration with **transport planning** for delivery scheduling and warehouse capacity.
- Automated **customs import filings** and built the tariff and duty-rate services used to price imported goods.
- Delivered core parts of the **About You integration**: made purchasing rules configurable instead of hard-coded, and co-built the system for internal sales and stock-ownership transfers between the two companies.
- Delivered **Spring Boot 3** migrations across inbound and procurement services and moved **12 services to Spring Boot 4**; extended **shared messaging libraries used by other teams** with tracing and event deduplication.
- Co-built **Shipping Notice Co-pilot**, an AI prototype that pre-fills supplier delivery notices from order history; the team won **“Best AI Usage”** at Procure-to-Pay HackWeek 2026.
- Spoke on **“The AI-Native Engineer”** at an internal engineering meetup; selected as a **Gen AI Labs trainer** for AI agents and MCP; co-designing a hands-on workshop for **~20 engineers across three teams**.
- Represented the shipment domain in **cross-team architecture** discussions; onboarded **three engineers**, interviewed candidates, reviewed **400+ code changes** from teammates, and led incident root-cause analyses.

Tech: Java, Kotlin, Spring Boot, PostgreSQL, Kafka (Nakadi), Kubernetes, AWS, OPA/Rego, OpenTelemetry

Java Engineer

Oct 2021 – May 2022

DHL Freight Enterprise

Sofia, Bulgaria (Remote)

- Developed features for DHL Freight’s **Transport Management System (TMS)**, which runs customers’ end-to-end transport operations across DHL’s regional and national network.
- **Automated logistics workflows**, reducing manual data entry and human error in shipment scheduling.
- Built deep **freight-logistics domain knowledge** through DHL’s domain training, applied directly to TMS work.

Tech: Java, Hibernate, internal WexVs Designer/Client frameworks

Software Engineer

Jan 2021 – Jun 2022

Rubikal

Alexandria, Egypt

- **Redesigned the data pipeline**, cutting average ETL time by **45%** and enabling real-time analytics.
- Analyzed education data covering **millions of students** and delivered actionable insights to clients.
- Built **Tableau and Metabase dashboards** that presented findings clearly to client stakeholders.
- Applied the **GTFS** transit-feed standard so public transit agency Mowasalat Misr could publish its transit data.
- Built and deployed serverless functions on **AWS Lambda**.

Tech: Spring, Rails, Python, SQL, Snowflake, dbt, AWS, Metabase, Tableau, Jira

Software Engineer

Apr 2020 – Dec 2020

Sia

Cairo, Egypt

- Built **7+ microservices** (gamification, notifications, educational content) for a 10-person ed-tech startup.
- Created an **end-to-end CI/CD pipeline** that cut deployment time by **60%**, enabling much faster feature releases.
- Optimized database queries, improving data-retrieval times by **30%** and speeding up responses for users.

Tech: Kotlin, Spring Boot, Angular, AWS, Docker, Jenkins, Jira

Founder, Freelance & Part-Time

Founder & CEO

Jan 2024 – Jan 2026

Noaat — Restaurant & retail tech startup

Alexandria, Egypt

- **Founded and scaled a B2B SaaS platform from 0→1**, serving **50+ restaurant clients** at **99.9% uptime**, and evolved it from a customer feedback and loyalty app into a cashback marketplace and a point-of-sale (POS) system.
- **Built and led a cross-functional team of 8 engineers** (frontend, mobile, backend, QA) plus data and design; set the technical roadmap, agile delivery (**40 two-week sprints**), code-review standards, and on-call playbooks, and was the main code reviewer (**330+ reviews**).
- **Architected the full technology stack**: Kotlin/Spring Boot backend (~**240 API endpoints**, **1,400+ tests**), React Native mobile app, Next.js website, and QR table ordering on AWS (ECS, Lambda, RDS, S3, CloudFormation).
- Designed a **multi-tenant architecture** supporting **50+ branches** with role-based access control, staff permissions, and centralized client management.
- Built an **enterprise point-of-sale system** with real-time order management, multi-payment support, kitchen display integration, and thermal printing through a custom Android printer bridge; personally built the POS web app (**1,500+ automated tests**, 40 releases).
- Implemented a **coupon and cashback loyalty system** with receipt-based claims and automatic receipt verification (AWS Textract, OpenAI), driving a **30% increase** in repeat customer visits.
- Built a **customer feedback platform** with survey creation and response analytics for data-driven improvements.
- Built an **Arabic AI sales and support chatbot** on Facebook Messenger (n8n, GPT-4o-mini, Pinecone) that answers customers, opens Jira tickets for issues, and routes sales leads to the CRM and Slack.
- Led a cloud security and reliability overhaul: moved the AWS setup to version-controlled infrastructure, **found and fixed a 14-month silent data-pipeline outage**, ran incident response, blocked an SMS-verification flood attack, and scaled the backend horizontally.
- **Drove product-market fit** through direct client engagement, iterating on features and closing enterprise accounts.

Tech: Kotlin, Spring Boot, PostgreSQL, Redis, TypeScript, React, React Native, Next.js, AWS (ECS, RDS, Lambda, S3, CloudFront, CloudFormation, Textract), OpenAI, n8n

Senior Backend Engineer (Part-Time)

Oct 2021 – Mar 2026

Cynopsis Solutions — RegTech (AML/KYC)

Singapore (Remote)

- Scaled AML/KYC transaction-processing infrastructure to **1M+ daily transactions**, improving throughput by **300%** through architecture optimization.
- Designed and implemented **20+ configurable compliance rules** covering transaction velocity, structuring detection, cross-border flows, and watchlist screening.
- Led the AI integration strategy, building a **RAG-powered compliance assistant** that cut analyst investigation time by **60%** and enabled natural-language regulatory queries.
- Directed **platform modernization**: multi-tenant architecture, CRM integration for user provisioning, and audit-ready permission frameworks.
- Designed a **case-workflow engine** with configurable status transitions, escalation paths, and ageing calculations for **regulatory SLA compliance**.
- Led **internationalization** with Arabic right-to-left support, enabling expansion into **Middle East markets**.
- Built an **executive Metabase dashboard** with real-time analytics on pending-case distribution, risk bucketing, and analyst performance.
- Authored **API documentation and technical design documents**, speeding up onboarding and third-party integrations.

Tech: Python, Django, LangChain, Celery, TimescaleDB, PostgreSQL, vector databases, Metabase

Cloud Native Engineer

Oct 2020 – Feb 2021

DHL Freight Enterprise

Cairo, Egypt

- Led the migration of monolithic applications to a **serverless architecture**, improving scalability and cutting operational costs by **35%**.
- Rebuilt a legacy PHP/Laravel codebase as **Azure Functions** with Entity Framework, reducing latency by **40%** without breaking existing integrations.
- Automated **GitLab CI/CD** deployment pipelines, cutting deployment time by **60%** and reducing production errors.
- Optimized **Cosmos DB** data access, speeding up retrieval by **25%** while maintaining high availability.

Tech: Azure Functions, Entity Framework, Cosmos DB, Laravel, GitLab CI/CD, Serverless

Software Engineer

Aug 2018 – Apr 2020

Robo-Tech

Alexandria, Egypt

- Built the communication system for **remotely operated vehicles (ROVs)**, reducing packet loss to below **0.01%**.
- Designed and launched **2 ROV prototypes** that passed testing in varied underwater conditions and won client approval for mass production.
- Improved real-time object-detection accuracy by **30%** by optimizing the vehicle's **OpenCV**-based navigation system.
- Cut prototype delivery time by **20%** by shortening development cycles with a cross-functional engineering team.

Tech: C++, Qt, C#, OpenCV

Technical Skills

Languages: Java, Kotlin, Python, TypeScript, SQL, C++, Ruby

Backend: Spring Boot, Django, Node.js, Rails, Celery, Testcontainers

Data: PostgreSQL, Redis, MongoDB, TimescaleDB, Snowflake, dbt

Architecture: Microservices, event-driven architecture, Kafka (Nakadi), REST API design, OPA/Rego, multi-tenancy

Cloud & Operations: AWS, CloudFormation, Azure, Kubernetes, Docker, CI/CD, OpenTelemetry, Grafana

AI & Agentic Engineering: Claude Code, GitHub Copilot, MCP, LangChain, RAG, vector databases, embeddings

Frontend & Mobile: React, Next.js, React Native, Angular

Education

Alexandria University, Faculty of Engineering

Graduated 2020

Bachelor of Science in Computer & Communication Engineering

Alexandria, Egypt